Design and Analysis of Compact Microstrip Resonator-Base Bandpass Filter for wireless Communication Applications

M. Naveen Kumar
Dept. of ECE
PSCMR College of Engg. & Tech.
Vijayawada, AP, India
mnaveenkumar@pscmr.ac.in

K.Lokesh Gangadhara Rao
Dept. of ECE
PSCMR College of Engg. & Tech.
Vijayawada, AP, India
kaapavarapul@gmail.com

K.Madhavi Sri Vaishnavi
Dept. of ECE
PSCMR College of Engg. & Tech.
Vijayawada, AP, India
madhavisrivaishnavi@gmail.com

M.Jyothi Swaroop
Dept. of ECE
PSCMR College of Engg. & Tech.
Vijayawada, AP, India
makinajyothiswaroop@gmail.com

P. Krishna Sanjay
Dept. of ECE
PSCMR College of Engg. & Tech.
Vijayawada, AP, India
Sanju.pothana@gmail.com

***Abstract*—This paper presents the design and implementation of a Band-pass filter (BPF) for ISM band wireless communication applications. This Bandpass filter uses a modified resonator structure with optimized coupling and compact microstrip layout to improve the filter performance.Conventional band-pass filters implemented on FR-4 substrates typically exhibit higher dielectric losses, limited selectivity, and restricted bandwidth. To address these limitations, the proposed design employs a Rogers RT/Duroid 5880 substrate, known for its low dielectric loss and suitability for high-frequency microwave circuits. The filter is modeled and analyzed using ANSYS HFSS simulation software.The use of an optimized substrate significantly enhances signal transmission characteristics, reduces power dissipation, and improves overall filter efficiency. The proposed design is well-suited for RF front-end modules, wireless transceivers, and IoT-based communication systems operating in the ISM band. The results demonstrate that appropriate substrate selection plays a crucial role in achieving high-performance and reliable microwave filter designs.**

***Index Terms*—Band-pass filter (BPF), U-shaped coupling, ISM band, Rogers RT/Duroid 5880, microwave filter, RF front-end, wireless communication, HFSS**

I. Introduction

Ring resonators are widely used in high-frequency systems and play a significant role in microwave, millimeter-wave, and optical applications due to their resonance characteristics. These structures operate based on the interaction of electromagnetic waves within a closed-loop geometry, resulting in constructive and destructive interference at specific frequencies. Owing to their ability to achieve high quality factor (Q) and efficient energy storage, ring resonators are commonly employed in the design of filters, sensors, oscillators, and frequency selective surfaces (FSS) [1]– [5].

Among their various applications, ring resonators are particularly suitable for band-pass filter design because they offer low insertion loss and compact size. By carefully controlling the resonator dimensions and coupling mechanisms, important parameters such as bandwidth, transmission response, and reflection characteristics can be optimized. However, conventional ring resonator designs often face challenges such as limited bandwidth, reduced selectivity, and increased structural complexity when multiple resonators or advanced geometries are used.

Several studies have explored different resonator configurations to improve performance. For example, dual-mode and multi-ring resonator structures have been proposed to enhance filtering characteristics, while split-ring and complementary split-ring resonators have been utilized for sensing and metamaterial applications. Although these designs achieve specific improvements, they often introduce design complexity, narrow bandwidth, or increased substrate requirements.

In addition, traditional microstrip-based filters fabricated on commonly used substrates may suffer from higher dielectric losses and degraded performance at higher frequencies. This limitation necessitates the development of improved resonator structures and the use of low-loss materials to achieve better efficiency and reliability.

To address these challenges, this work proposes a hybrid ring resonator structure that combines square and circular ring geometries with U-shaped coupling elements. The proposed configuration enhances mutual coupling, improves bandwidth, and maintains a compact design. Furthermore, the use of an optimized low-loss substrate contributes to better signal transmission and reduced power dissipation.

The main objectives of this work are to design an efficient hybrid ring resonator with improved transmission characteristics, minimize losses, and achieve stable performance suitable for ISM band wireless communication applications. The proposed design also aims to provide a practical solution for RF front-end systems requiring compact size, high efficiency.

II. Design Methodology

The proposed band-pass filter (BPF) is designed using a compact microstrip resonator structure with improved coupling and reduced losses. The overall design procedure is carried out in multiple stages, as illustrated in Fig. 1 and Fig. 2. The methodology focuses on achieving compact size, stable frequency response, and efficient signal transmission.

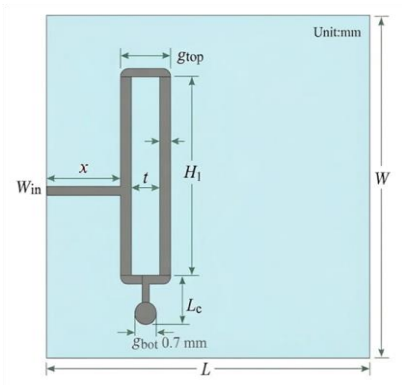

Fig. 1. Initial resonator structure (Step-1)

In the first stage, a single U-shaped resonator is designed, as shown in Fig. 1. The input feed line is connected to the resonator, and key parameters such as width, length, and coupling gap are selected to establish the fundamental resonance. This stage ensures proper impedance matching and basic filtering characteristics.

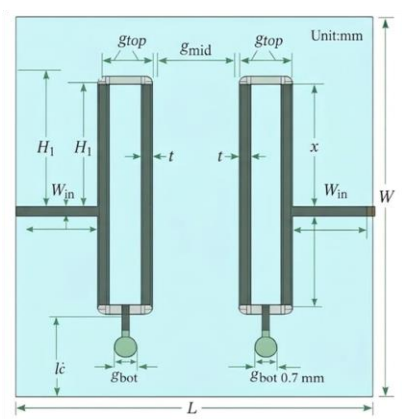

Fig. 2. Extended resonator structure with multiple units (Step-2)

In the second stage, additional resonator elements are introduced, as shown in Fig. 2. Multiple identical resonators are arranged in a cascaded configuration to enhance mutual coupling and improve bandwidth. Edge coupling between adjacent elements provides better control over frequency selectivity and transmission performance.

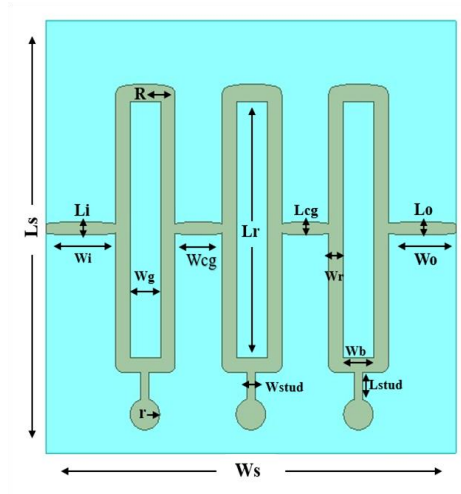

Fig. 3. Final compact microstrip resonator band-pass filter design

The final optimized design is presented in Fig. 3. The structure is implemented on a Rogers RT5880 substrate to reduce dielectric losses and improve high-frequency performance. The geometrical parameters are carefully tuned to achieve the desired operating frequency and bandwidth.

TABLE I
Design Parameters of Proposed Filter

Parameter	Value
Ls	10 mm
Ws	10 mm
Lcg	0.7 mm
Wcg	1.15 mm
Lr	6.3 mm
Wr	0.35 mm
Li	1.05 mm
Wi	1.7 mm
Lstub	0.7 mm
Wstub	0.2 mm
R	0.68 mm
r	0.36 mm
Lo	1.05 mm
Wo	0.35 mm
Wg	1.11 mm
Substrate	Rogers RT5880

The comparison between the existing and proposed designs is summarized in Table II. The proposed filter achieves reduced size, improved transmission characteristics, and better suitability for RF applications.

TABLE II
Comparison of Existing and Proposed Design

Parameter	Existing	Proposed
Resonator Type	Hybrid ring	U-shaped resonator
Structure	Square + circular	Repeated units
Coupling	U-shaped	Edge coupling
Size	16×70 mm	10×10 mm
Frequency	2.45 GHz	~3.5 GHz
S11	-30 dB	-58 dB
S21	-2.12 dB	Near 0 dB

Overall, the proposed methodology provides an efficient and compact solution for designing low-loss band-pass filters suitable for modern wireless communication systems.

III. Results and Discussion

A. Filter Design and Experimental Setup

1) Front View of Filter: The front view of the proposed filter is shown in Fig. 4. The filter is fabricated on a dielectric substrate with a compact microstrip structure. The radiating element consists of a modified patch geometry designed to achieve multiband characteristics. The feed line is connected through an SMA connector to ensure proper impedance matching. The geometry is optimized to enhance radiation efficiency and bandwidth performance.

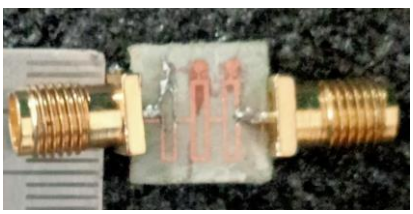

Fig. 4. Front view of the proposed Filter

2) Back View of Filter: The back view of the filter is illustrated in Fig. 5. It shows the ground plane configura- tion, which plays a crucial role in determining the antenna performance. A partial/modified ground structure is employed to improve impedance bandwidth and radiation characteristics. The ground plane dimensions are carefully chosen to support stable operation across the desired frequency range.

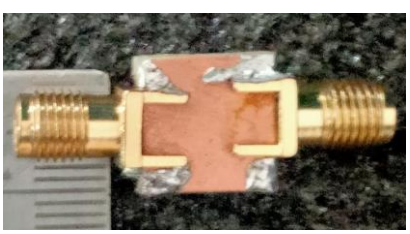

Fig. 5. Back view of the proposed Filter

3) Experimental Setup: The experimental setup used for antenna measurement is shown in Fig. 6. The filter is tested inside an anechoic chamber to minimize external interference and reflections. A microwave analyzer (Vector Network Analyzer) is used to measure key parameters such as return loss (S_{11}) and frequency response. The antenna is connected via coaxial cables with SMA connectors to ensure accurate signal transmission. The setup ensures reliable and precise characterization of filter performance.

Fig. 6. Experimental setup for filter measurement

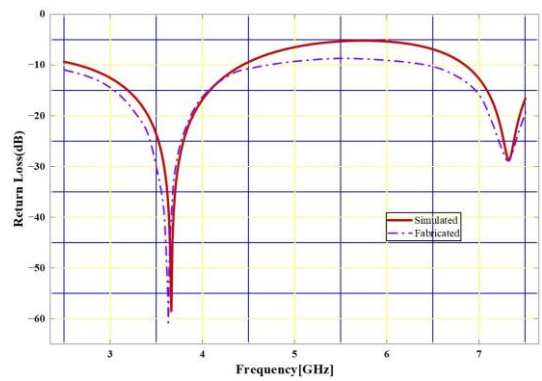

Fig. 7. Return loss (S11) of the proposed filter

Fig. 7 shows the return loss (S11) characteristics of the proposed band-pass filter. Return loss represents the amount of power reflected due to impedance mismatch. A lower S11 value indicates better matching between the input and the filter.

From the figure, it can be observed that the S11 value reaches below -58 dB around the operating frequency, indicating acceptable impedance matching. A deeper notch (around -50 dB near resonance) confirms strong resonance behavior and minimal signal reflection. This demonstrates that the designed filter effectively minimizes return losses and ensures efficient power transfer.

The performance of the proposed band-pass filter (BPF) is evaluated using ANSYS HFSS simulations. Key parameters such as transmission characteristics (S21), voltage standing wave ratio (VSWR), field distributions, and impedance matching are analyzed to validate the effectiveness of the design.

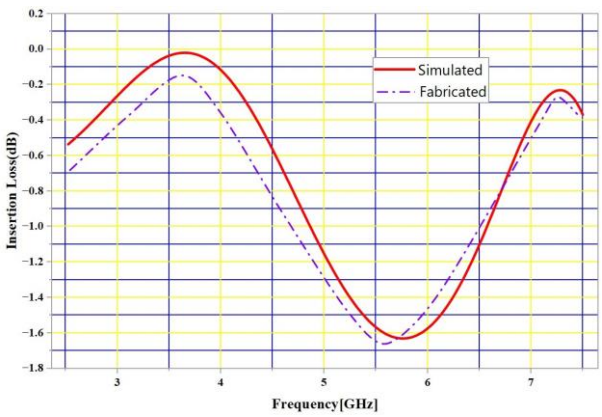

Fig. 8. Transmission coefficient (S21) of the proposed filter

Fig. 8 illustrates the transmission coefficient (S21) of the designed filter. It can be observed that the filter exhibits high transmission near 3.5 GHz, indicating efficient signal propagation through the structure. The passband is approximately in the range of 2.7 GHz to 4.4 GHz, demonstrating the capability of the filter to operate effectively within the desired frequency range.

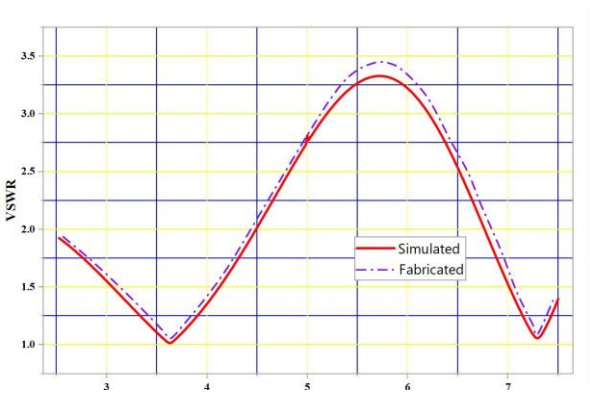

Fig. 9. VSWR characteristics of the proposed filter

The VSWR performance is shown in Fig. 9. At the operating frequency, the VSWR value remains below 2, which indicates good impedance matching and minimal signal reflection. This confirms that the designed filter allows efficient power transfer with reduced losses.

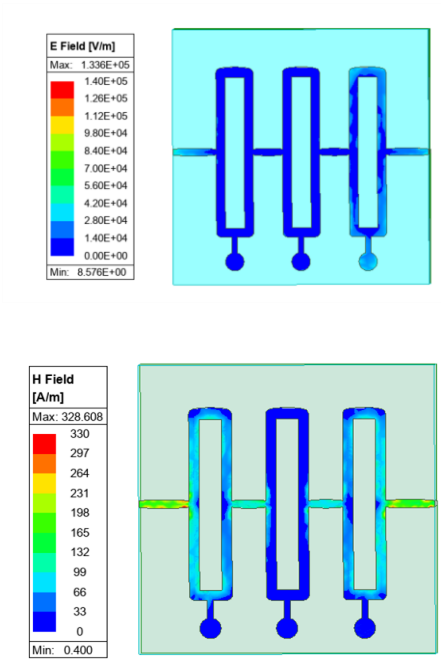

Fig. 10. Electric field (top) and magnetic field (bottom) distributions

Fig. 10 presents the electric (E-field) and magnetic (H-field) distributions within the resonator structure. The concentration of fields along the resonating elements confirms effective excitation and proper energy confinement. This behavior indicates that the resonator efficiently stores and transfers electromagnetic energy.

The impedance locus is located near the center of the chart around the operating frequency, confirming proper impedance matching and stable filter performance.

In addition to simulation results, important performance parameters such as bandwidth, center frequency, and quality factor are evaluated. The bandwidth is calculated as the difference between the upper and lower cutoff frequencies:

$$BW = f_H - f_L = 4.4\,\text{GHz} - 2.7\,\text{GHz} = 1.7\,\text{GHz}.$$

The center frequency is determined using:

$$f_c = \frac{f_H + f_L}{2} = \frac{4.4 + 2.7}{2} = 3.5\,\text{GHz}.$$

The quality factor (Q) of the filter is given by:

$$Q = \frac{f_c}{BW} = \frac{3.55}{1.7} = 2.1.$$

These results indicate that the proposed filter achieves a balanced performance in terms of bandwidth, selectivity, and transmission efficiency. Overall, the simulation outcomes confirm that the designed filter is suitable for RF and wireless communication applications requiring compact size and reliable performance.

IV. Conclusion

In this work, a compact microstrip band-pass filter based on a hybrid resonator structure has been successfully designed and analyzed. The proposed design demonstrates effective performance in terms of impedance matching, transmission efficiency, and bandwidth characteristics, making it suitable for modern RF and wireless communication applications.

The simulation results show that the return loss (S11) remains below -58 dB at the operating frequency, indicating good impedance matching and minimal signal reflection. The transmission response (S21) confirms efficient signal propagation within the desired passband range of 2.7 GHz to 4.4 GHz. Additionally, the VSWR value remains below 2, ensuring proper power transfer and reduced losses.

Field distribution analysis further validates the design by showing strong electric and magnetic field confinement within the resonator structure, indicating effective energy storage and propagation. The Smith chart analysis also confirms that the impedance is well-matched near the center of the chart at the operating frequency.

Furthermore, the calculated bandwidth of 1.7 GHz and center frequency of 3.5 GHz demonstrate that the filter achieves a balanced trade-off between selectivity and bandwidth. The quality factor (Q) value indicates satisfactory filter performance for practical applications.

Overall, the proposed filter design offers a compact size, low loss, and reliable performance, making it a suitable candidate for RF front-end modules, wireless communication systems, and IoT applications. Future work may focus on experimental validation and further optimization for multi-band or reconfigurable designs.

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